

CostVision

AI-Powered Should-Cost Intelligence for Modern Engineering

Designed & Developed by Avinash Bhosale

Cost Engineering & Digital Innovation

21 Commodity Models

AI Agent (NLP)

CAD-to-Cost AI

DFM / DFA Layer

20 Regions · 10 FX

Assembly BOM Rollup

Wright's Law Learning Curve

Supplier Quote Comparison

Cloud Team Sync

THE CHALLENGE

Why CostVision? Traditional Costing is Broken

Manual effort, weeks of delay, and supplier dependency cost organisations millions annually.

 Weeks, Not Minutes

Manual should-cost takes 2-4 weeks per part. Sourcing decisions stall. Programme timelines slip.

 Supplier Dependency

Without independent cost intelligence, engineers accept quotes at face value — no leverage, no transparency.

 Inconsistent Methodology

Every engineer calculates cost differently. No standard model. No auditable assumptions. No cross-team comparability.

 No Regional Visibility

Labour, machine, material and energy rates vary dramatically by country. Manual adjustment is error-prone.

 Late-Stage Cost Surprises

Cost issues discovered at SOP — not at concept. DFM problems missed. Expensive redesigns. Margin erosion at launch.

 No Supplier Quote Benchmark

Buyers lack a defensible floor price for negotiation. Supplier margins are opaque and unchallenged.

80%

of part cost is locked in at the design stage — yet most cost analysis happens after design freeze. CostVision shifts cost intelligence to where it matters: concept phase.

THE SOLUTION

What is CostVision?

A world-class should-cost engine with AI at every layer — from natural language input to DFM/DFA output.

 **AI Agent (Natural Language)**

Describe a part in plain English. AI auto-builds the complete cost model: material, routing, machines, operations.

 **CAD-to-Cost Automation**

Upload STEP / IGES / photo. AI extracts geometry, infers material, selects process model — full cost in minutes.

 **21 Engineering-Grade Cost Models**

Physics-based models for every major commodity. Routing logic, yield curves, tooling amortisation — all encoded.

 **Assembly BOM Rollup**

Cost entire assemblies — not just single parts. Multi-level BOM with per-component commodity models.

 **Learning Curve / Wright's Law**

Project cost at any volume milestone. Configure learning curve %. Essential for LTA negotiations.

 **Supplier Quote Comparison**

Log supplier quotes. Compare vs should-cost instantly. Identify margin gaps. Generate negotiation talking points.

 **DFM / DFA Intelligence**

Bolt-on AI scores manufacturability 1–10. Critical issues, saving potential, actionable recommendations.

 **20 Regions · 10 Currencies**

Real-time labour, machine, material and energy rates. Currency auto-switches when region is selected.

 **Export, Scenarios & Team Sync**

6-sheet Excel, PDF report, A/B/C scenario comparison, tornado sensitivity chart, cloud team sync.

GAME-CHANGING CAPABILITY

AI Agent — Natural Language Should-Costing

The fastest path from part concept to full cost breakdown: just describe the part in plain English.

How It Works

1. Describe

Type a part description: 'Aluminium bracket, 6082-T6, approximately 200×100×50mm, 3 machined holes, anodised finish, 5,000/year UK production'

2. AI Analyses

Claude AI interprets the description, infers geometry, selects the most cost-effective manufacturing route, and chooses appropriate machines and labour grades

3. Auto-Populate

The cost model is automatically populated: material weight, 3 machining operations, cycle times, OEE, tooling NRE — all filled without manual input

4. Full Result

Click Calculate. Full 8-bucket cost breakdown, AI insights, DFM/DFA score, sensitivity analysis, and export — all ready in under 60 seconds

● CostVision AI Agent

You: Describe a 4-layer FR4 PCB, 100×80mm, ENIG finish, flying probe test, 10,000/yr, UK manufacture

AI: Got it. Analysing your PCB specification...

- 4-layer FR4 standard, 100×80 mm board
- ENIG surface finish (+22% vs HASL)
- Flying probe electrical test
- UK pricing region applied
- NRE amortised over 10,000 boards

You: What is the should-cost?

AI: Should-Cost Result: £ 4.18 per board

- PCB Fab (material + process): £ 3.24
- Test (flying probe): £ 0.71
- NRE amortised: £ 0.23

DFM Score: 8.5/10 — Good manufacturability

Key Benefits ⚡ Zero form-filling 🎯 Any commodity ✓ Assumptions visible 📄 Full audit trail

AUTOMATION

CAD-to-Cost: Geometry to Should-Cost Automatically

Upload a STEP file or part photo — AI extracts features, infers material, and generates a full cost model.

<0.01%

Engine vs hand calc,
reference part

8

Cost buckets, each traced
to a rate

6

Real parts pinned in
regression

Supported formats

.STEP .IGES .STP .IGS .STL .JPG .PNG .HEIC

Drag-drop or file picker — CAD geometry measured by a full kernel, photos read by vision.

What AI Extracts from CAD / Photo

- Part geometry: envelope dimensions, volume, wall thickness, surface area
- Feature recognition: holes, pockets, ribs, undercuts, threads, datum features
- Material inference: visual texture & density → grade selection (e.g. AI 6082-T6)
- Process selection: geometry → most cost-effective manufacturing route
- Cycle time: material removal rate, tool path length, fixture & setup strategy
- Complexity: Simple / Moderate / Complex → machine type & OEE selection
- DFM pre-check: thin walls, tight radii, undercuts flagged before costing begins

AI Analysis Result — Aluminium Bracket

AI Detected Material
AI 6082-T6

Envelope (L×W×H)
220 × 140 × 85 mm

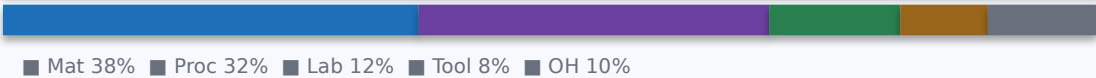
Complexity
Moderate — 3 setups

Suggested Process
3-Axis CNC Milling + Drilling

Est. Net Weight
0.82 kg

DFM Pre-check
1 flag: 3 setups → consolidate

Should-Cost Preview **£ 28.40**



NEW IN 2026 · ENGINEERING-GRADE VIEWER

3D CAD Viewer — Inspect, Measure & Analyse

A SaaS-grade CAD viewer built into CostVision — STEP / IGES / STL, in the browser, no separate CAD seat.

Views & Navigation

Iso / Front / Top / Right, Fit, orientation view-cube, smooth logarithmic-depth zoom (no z-fighting on metre-scale parts), maximize / full-window.

Display & DFM Analysis

Shaded / wireframe, bounding-box dims, grid toggle, colour-by-face-type, per-component colours + legend, draft & undercut heatmap, wall-thickness heatmap.

Structure & Assembly

Collapsible model tree (bodies · features · faces), section planes X/Y/Z, exploded view (Radial / X / Y / Z), per-part Move & Rotate — isolated, the rest of the assembly stays put.

Measure & Inspect

Distance, radius / diameter, angle, point X/Y/Z, face-to-face perpendicular — exact B-rep readouts, CSV export, snapshot straight into the cost report.

SaaS-grade UX

Docked collapsible tree, drag-to-reorder tool groups, slide-away toolbar, professional line-icon set — fits cleanly beside the app sidebar at any width.

One viewer, many jobs

24 tools · 6 groups
STEP / IGES / STL
Runs in the browser
B-rep-accurate geometry

Everything measured here
feeds the cost engine →

CAD-TO-COST · ZERO MANUAL ENTRY

What's Auto-Captured — Fed to the Cost Engine

Geometry is measured, not typed. AI only classifies material & process; every £ is deterministic and traceable.

Auto-captured from geometry (no typing)	→ Feeds this cost driver
Volume — exact, from the CAD kernel (OCCT)	Material mass · stock size
Weight per material = volume × density	Raw-material £ (Al / steel / iron / Cu / Ti)
Bounding box / envelope	Machine sizing · stock · setups
Body / component count	Assembly cost · BOM line count
Hole & boss table — Ø, depth, count	Secondary machining: drill / bore / tap
Wall thickness — min / mean / heatmap	Moulding & casting feasibility · cooling
Draft & undercut analysis	Tooling complexity — slides for undercuts
Face-type areas (planar / cylindrical)	Machining area · paint / coat area
CNC cycle-time & setup estimate	Process-time baseline

⚙️ Worked example — Spur Gear (m3 · z38)

20MnCr5 case-hardening steel — captured automatically:
Volume 264.9 cm³ · Weight 2.08 kg
Envelope 120 × 120 × 30 mm · 1 body

→ Should-cost, India ex-works (GBP):
Material £3.36 · Process £11.40 · Labour £1.36
Overhead £1.95 · Margin £1.83 + heat-treat £1.77
= £21.90 / part (≈ ₹2,398) · £10.53 / kg

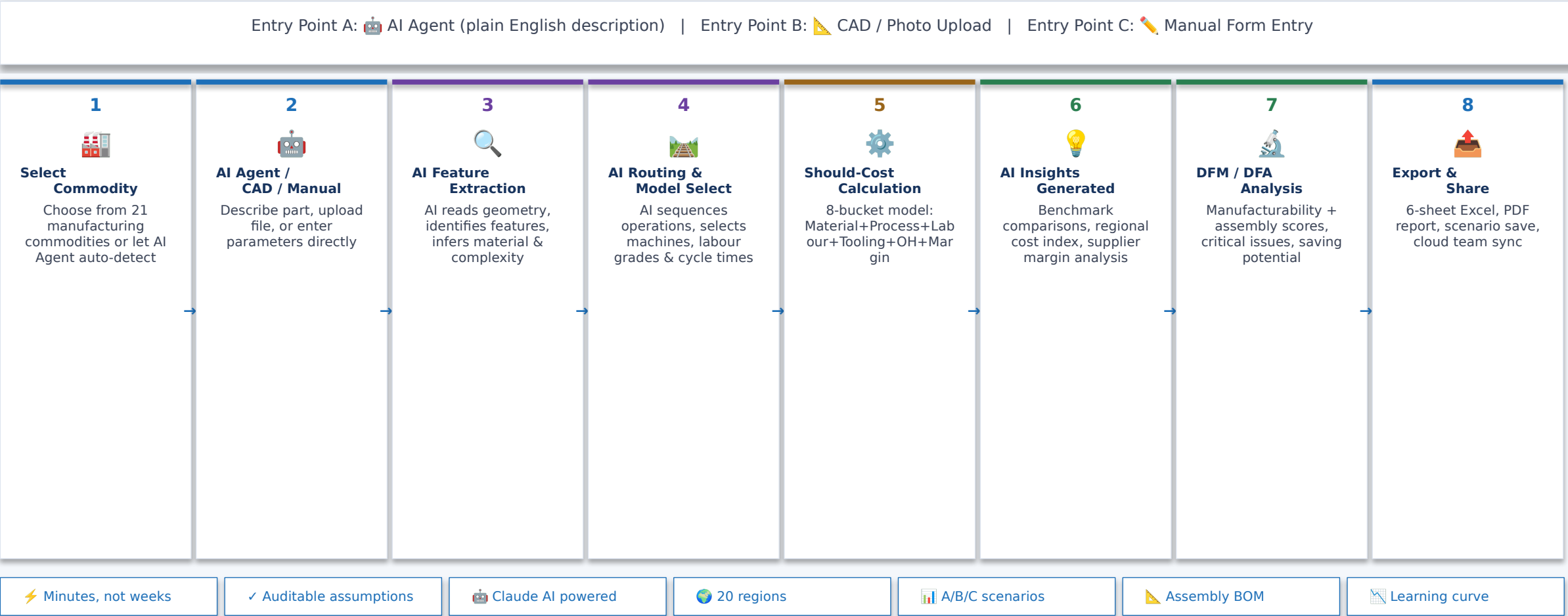
🔒 The golden rule

AI never sets a price. It only reads and classifies — material family, process route, feature intent. Every number above is deterministic arithmetic in the cost engine, bounded by the measured geometry and traceable to the rate library. That is what makes the output defensible.

WORKFLOW

AI Agent — 8-Step End-to-End Flow

Three entry points. One intelligent engine. Full cost intelligence in under 5 minutes.



FULL COVERAGE

21 Manufacturing Commodities — One Platform

Every major manufacturing process — all modelled to engineering depth with full routing, yield, and tooling logic.

 CNC Machining 3/4/5-axis, turning, grinding	 Casting HPDC, gravity, investment, sand	 Cast + Machine Casting + post-ops + HT	 Forging Hot, warm, cold die forging	 Sheet Metal Progressive/transfer die	 Sheet Metal Fab Laser · punch · press brake · MIG	 Gear Cutting Hob · shape · skive · broach · grind
 Injection Moulding Thermoplastics, multi-cavity	 Blow Moulding HDPE, PET, extrusion blow	 Extrusion Profile, pipe, sheet, rod	 Thermoforming Vacuum, pressure, twin-sheet	 Rotational Moulding Large hollow shapes, LLDPE	 Rubber Moulding Compression, injection, LSR	 Composites CFRP/GFRP, autoclave, RTM, AFP
 Painting / Coating E-coat, powder, wet, anodise	 BIW Assembly Spot weld, MIG, hem, bond	 PCB Fabrication 2-16L, HDI, rigid-flex, RF/μwave	 PCBA / SMD SMT, TH, reflow, conformal coat	 Wiring Harness Cut-strip-crimp, test, sub-asm	 Assembly BOM Rollup Multi-part BOM, full cost rollup	 AI Agent Natural language → any commodity

ENGINEERING DEPTH

What Sits Behind Every Number

Twelve layers, all of them arithmetic you can follow. Nothing is estimated by feel.

 Materials 30+ grades — steels, aluminium, plastics, rubber, composites, board laminates. Priced in the local currency.	 Machines & Rates 50+ machine types with an hourly rate you can edit per project and per region.	 Process Routing The order of operations, and for each one: which machine, how long, who runs it.	 Tooling & NRE Dies, moulds and fixtures spread over the year's volume, with tool life accounted for.
 Yield & Scrap What actually comes out good first time, driven by the features that make a part hard to build.	 Regional Rates 20 regions, each with its own labour, energy and machine rates.	 Benchmarks & Sensitivity Checked against published cost ranges, and ranked by which inputs move the number most.	 Learning Curve What the part costs as volume builds — the figure you need for a long-term agreement.
 Quote Comparison Put the supplier's quote next to ours and see the gap, in their currency or yours.	 Stated Assumptions Where something isn't known, the AI fills it in and says so — every assumption is on the page.	 Gear Cutting Hobbing, shaping, skiving and grinding times worked out from the gear itself, not a rule of thumb.	 Landed Cost Duty, carbon levy and shipping terms — the cost at our door, not at the supplier's gate.

ADVANCED CAPABILITIES

Beyond a Cost Number

Five tools that turn the estimate into something you can negotiate with.

 Learning Curve

Cost falls as volume builds. Set the curve, read the cost at any year.

- Quote year-three cost, not prototype cost
- Built for long-term agreements and make-vs-buy

 Assembly Roll-Up

Cost a whole module, not just one part.

- Mix any of the 21 commodities in one build
- Every component keeps its own cost model
- Body + paint + harness + PCBA = one total

 Quote Comparison

Log the supplier's quote. Any currency, converted for you.

- Tells you the gap in words: "18% above should-cost"
- Gives you the talking points to open with

 What Moves the Number

Every driver flexed $\pm 10\%$, ranked by impact.

- Shows the few inputs worth checking before you negotiate
- Chart exports straight into your supplier pack

 Scenarios Side by Side

Baseline, target and stretch on one screen.

- See exactly where the money moved
- Save and share with the team

AI INTELLIGENCE LAYER

DFM / DFA — How to Make the Part Cheaper

Runs after the cost is settled. It reads the result and advises — it never changes the price.

Manufacturability Score (DFM)

8.5 / 10

Assembly Efficiency (DFA)

7.0 / 10

Total Potential Saving

~19.3%

Scored out of 10 · saving is the top three fixes combined

 What It Finds — Making the Part

Too much material going to scrap.
Machines running well below what they should.
Tooling, overhead or supplier margin out of line.
Each issue graded critical, major or minor.

 What It Finds — Building It

Assemblies done by hand that a robot could do.
Too many setups — every one adds cost and variation.
Too many different fasteners, so tools keep changing.

 What to Do About It

Six to eight actions per part, with the saving on each:

Automate the manual operations 5-20%
Move to a near-net shape 5-15%
Raise machine uptime 5-12%
Plus quick wins: more volume, open-book overhead, re-bid.

 Ask the AI to Go Deeper

One click, after the rule-based findings are on screen.

Why the cost sits where it does.
The three things to fix first.
What to say to the supplier, and a different process to consider.

GLOBAL COVERAGE

20 Manufacturing Regions — 10 Currencies

Select a region and every rate adjusts automatically — labour, machines, energy, material and currency.

Key Manufacturing Regions (sample)

Region	Currency	Sample Labour Rate	Position
 UK	GBP £	Skilled £26/hr	Industry / Aerospace baseline
 Germany	EUR €	Skilled €32/hr	Premium, high automation
 Poland	EUR €	Skilled €11/hr	Low-cost EU option
 Romania	EUR €	Skilled €7/hr	Ultra-low EU labour
 China	CNY ¥	Skilled ¥28/hr	Global volume manufacturing
 India	INR ₹	Skilled ₹320/hr	Engineering & electronics hub
 Mexico	MXN \$	Skilled MXN82/hr	Near-shore for North America
 USA	USD \$	Skilled \$32/hr	Premium North American rates
 Thailand	THB ฿	Skilled ฿180/hr	Electronics & automotive
 Vietnam	VND ₫	Skilled ₫52k/hr	Lowest-cost electronics hub
 Brazil	BRL R\$	Skilled R\$22/hr	South America manufacturing
 S. Korea	KRW ₩	Skilled ₩22k/hr	Electronics & automotive

Auto-Currency Switch

When you select a manufacturing region (e.g. China), the display currency automatically switches to CNY. No manual FX conversion needed.

Full Rate Adjustment

Every rate adjusts: skilled/semi-skilled/engineer/inspector labour, electricity, gas, machine rates, material prices — all recalculated for the selected region.

Editable Rate Library

All rates are editable. Override any rate with project-specific data, supplier quotes, or latest market prices. Resets to calibrated baseline at any time.

Regional Comparison

AI Insights tab shows estimated cost in all 20 regions simultaneously — instant make-vs-buy regional arbitrage analysis for any part.

THE CONCEPT • EXPLAINED

Agentic vs Autonomous Agentic AI — With Examples

Two levels of "agentic": on the left the AI acts because you asked; on the right it acts on its own, unattended.

Agentic AI

Assisted — you trigger it and stay in the loop; a human approves

AI Agent

Describe a part in plain English — it builds the full cost model for you.

CAD / photo → cost

Upload a STEP file — it measures geometry and infers material & process.

Negotiation coach

Open a part — it drafts the buyer's counter-argument and target price.

Rate-data assistant

Ask a costing question — it answers from your own rates, with citations.

Autonomous Agentic AI

Self-directed — runs unattended on a schedule, within limits you set

Savings monitor

Runs on the server, compares paid vs should-cost, opens findings itself — £0.5M/yr unattended.

Self-audit

Re-checks every estimate for known errors and corrects within bounds — unasked.

Calibration & drift

Learns from logged quotes, re-derives factors, watches for drift continuously.

Outcome-weighted ranking

Learns which findings actually convert and re-prioritises its own queue.

The common thread: you set the boundaries and every action stays glass-box and auditable — autonomy never means the AI sets a price in secret.

NEW IN 2026 · AGENTIC INTELLIGENCE

The Tool Now Checks Its Own Work — and Learns

It re-checks every estimate before you see it, and it learns from the quotes you have already had.

 It checks its own homework

Every estimate is re-checked against the mistakes this tool has actually made before — wrong process, impossible machining time, a weight that disagrees with the CAD. It corrects them and shows its working. The measured geometry always wins over the AI.

 It learns from your quotes

Load the supplier quotes you already have. The engine corrects itself against them, and the estimate gets tighter as your history grows. Every estimate now says how well calibrated it is.

 It picks the right machine

One rule now sizes the machine for the part on every commodity — presses, die-casting, moulding, extrusion — instead of a fixed guess per commodity.

Proven on five real automotive CAD parts

Each was mis-costed on the first pass; the self-audit caught it and corrected it.

Real Part	AI First Pass	What Was Wrong	Corrected
Fuel tank	£216.97	Treated as cast	£24.62 blow-mould
Front bumper	£29.18	Machined, not moulded	£7.79 injection
Servo horn	£333.00	Bulk, not net-shape	£6.40 net-shape
Stub axle	no result	Classed as the wrong process	Forged steel 8.1 kg
Seat cross-member	no result	Machine sizing gap	£1.11 runs clean

Built to be trusted, not just believed

- 1,777 automated tests — if the logic breaks, the build fails, not the demo
- Every estimate is re-checked before you ever see the number
- The CAD engine is tested in the same container we deploy

LIVE DEMO DATA

Real-World Cost Benchmarks — Luxury SUV Programme

All figures generated by CostVision's should-cost engine. UK manufacturing region, GBP.

Commodity	Demo Part	Key Inputs	Should-Cost	AI Top Insight	DFM Flag
 Machining	AI Gearbox Bracket	Al 6082-T6 · 0.85 kg · 3 setups	£ 28.40	Process 44% → near-net-shape saves £8	Major: 3 setups → 5-axis
 HPDC Casting	Transmission Housing	Al A380 · 3.2 kg · 50k/yr	£ 18.70	Die 9% OK. Margin 21% → negotiate	Good — no critical issues
 Sheet Metal	Rear Door Inner	DC04 1.2mm · 4.1 kg · 60k/yr	£ 22.10	Blank nesting 64% → improve saves £2.80	Critical: nesting <65%
 Injection	PP Bumper Bracket	PP-TD20 · 0.42 kg · 100k/yr	£ 3.85	Mould 8% — good at this volume	Good — runner waste 15%
 Rubber	EPDM Door Seal	Extrusion · 2.8m · 85 Shore A	£ 4.10	Process 42% — cure cycle dominant	Major: cure time optimise
 PCB Fab	4-Layer FR4 ECU Board	100×80mm · ENIG · Flying probe	£ 4.18	NRE 6% per board — increase batch	Good — standard technology
 PCBA	ECU PCBA 85 comp.	SMT + TH · 4 TH joints	£ 32.60	Labour 38% — convert TH to SMT saves £4	Major: high TH labour
 Wiring Harness	Door Module Harness	24 circuits · 2.1m avg	£ 11.80	Labour 58% — automation saves 20%	Critical: labour >50%
 Composites	CFRP Floor Pan	Prepreg CFRP · 1.8 kg · autoclave	£ 185.00	Labour 44% — evaluate ATL/AFP	Major: manual layup cost
 Forging	Steel Control Arm	42CrMo4 · 2.1 kg · 80k/yr	£ 9.20	Die 11% — acceptable at volume	Good — closed-die efficient

REPORTING & COLLABORATION

Getting the Number Out of the Room

Reports people can act on, and one shared baseline for the whole team.

 Excel — Six Sheets

The whole build-up, not just the total.

Summary · material · every operation · machine rates · labour rates · and a sheet recording every assumption.

 PDF Report

The version you send out.

Cover with the part photo, the cost summary, the operation-by-operation build, and the savings and DFM findings.

 One Shared Baseline

Everyone works from the same numbers.

Saved centrally, not on a laptop.
No arguing over which spreadsheet is current.

 Part Photos

Attach a photo to any calculation.

It appears on the report cover — and the AI can cost from the photo itself.

 Secure Access

Enterprise sign-in, as you would expect.

Email verification, password reset, and sign-in attempts capped.

 Built-In Help

Nobody needs training to start.

Getting started, the AI agent, CAD and photo, a glossary and troubleshooting — all in the app.

VALUE DELIVERED

What This Is Worth

The same claims, in numbers a finance team recognises.

70-90%

Reduction in costing time

<0.01%

Engine vs hand calc, reference part

15-25%

Typical supplier price reduction

3x

Faster sourcing decisions

£15M

On £500M of spend, a 3% improvement

 Speed

Two to four weeks becomes ten minutes. Engineers spend the day deciding, not building spreadsheets.

 Accuracy

Built from the physics of the process, every rate on record. Accuracy is measured as real prices are logged — not claimed up front.

 Negotiation

A floor price you can defend, part by part. Supplier margin becomes visible instead of assumed.

 Early Warning

Most of the cost is fixed at design. Problems surface while changing them is still free.

 Where to Build It

Compare 20 regions in one click. Make-vs-buy answered with real rates.

 One Method

The same model everywhere, with every assumption on record. Leadership sees one set of numbers.

 The Right Volume

See what the part costs as volume grows, and when a new process starts to pay.

 Shared Knowledge

One baseline the whole team works from. Cost know-how stays with the business, not one engineer.

VISION & NEXT STEPS

Intelligent, Instant, Integrated — What Comes Next

CostVision is live today. Here is what comes next — and how to get started.

Current Capabilities — Live Today

- 21 commodities costed to engineering depth
- Describe a part in plain English — the model builds itself
- Upload a CAD file or a photo, get a cost in minutes
- Every estimate re-checked, and corrected, before you see it
- Learns from the supplier quotes you already have
- Whole assemblies, not just single parts
- Learning curve, quote comparison and scenarios
- 20 regions and 10 currencies, switched automatically
- Excel and PDF reports, shared across the team

Phase 2 — H2 2026

Connect to Teamcenter and SAP
Cost-target dashboard for programme teams
Compare several suppliers side by side

Phase 3 — 2027

Calibrate itself from live ERP actuals
Draft the RFQ straight from the should-cost
Portfolio view for the leadership team

Phase 4 — 2027+

Cost that updates as the design changes
AI suggestions for redesigning the part
Supplier risk scoring · mobile app for plant visits

Pilot Rollout — Next Steps

- Two or three commodities, one programme team
- Four to six weeks, on real parts against real quotes
- We call it a success at 15% price reduction found per part
- Avinash Bhosale — Cost Engineering & Digital Innovation